

Adi Roitburg

Seattle, WA | (206)-852-5861 | adi.roitburg@gmail.com | [linkedin.com/in/adi-roitburg](https://www.linkedin.com/in/adi-roitburg) | github.com/Adiro777

EDUCATION

University of Colorado Boulder

2026 – Present

M.S. in Computer Science

Loyola Marymount University

2022 – 2026

B.S. in Computer Science

Minor in Interactive, Gaming, and Immersive Media

- Relevant coursework: Data Structures and Algorithms, Algorithms and Analysis, Computer System Organization, Artificial Intelligence, Game Design, Game Development, Operating Systems, Cognitive Systems, Generative AI Algorithms
- GPA: 3.77 — Magna Cum Laude

WORK EXPERIENCE

Computer Science Teaching Assistant — Loyola Marymount University

- Taught Unity fundamentals to students in the Game Development course to support their class projects *Sept 2023 – May 2026*
- Supported students in the Beginner Computer Programming course with homework, projects, and debugging help

Machine Learning Researcher — Seattle University

June 2025 – Aug 2025

- Built a neural network in Python (TensorFlow, PyTorch) to predict DMSO concentration from dielectric measurements
- Ran GROMACS molecular dynamics simulations across varying water-to-DMSO ratios to model dielectric property changes and validate the model

PROJECTS

NLP to SQL Agent

Jan 2026 – May 2026

Python, LangChain, ChromaDB

- Built a LangChain ReAct agent that converts natural language questions into executable SQL queries
- Designed a RAG pipeline over ChromaDB to surface the relevant database schema for each query
- Implemented a read-only validation layer that rejects destructive operations before execution

Sniper Kitty 2D

Jan 2026 – May 2026

Unity, C#, Aseprite

- Built a 2D puzzle-platformer with character AI, a dynamic camera, and physics-based puzzle systems
- Developed a modular level system enabling scalable content creation from reusable components
- Implemented a cursor-based movement framework using input tracking to drive dynamic AI responses

Gravity

Jan 2025 – May 2025

Unity, C#, Aseprite

- Developed a 2D endless runner with a unique gravity-shifting movement system
- Built an extensible power-up/obstacle framework using shared base classes
- Implemented time-based difficulty scaling, progressively increasing obstacle spawn rate and speed

Robo Runaway

Sept 2024 – Dec 2024

Unreal Engine, C++

- Built a 3D parkour game with custom traversal systems (sprint, wall-run, chainable slide-jumps) for fluid movement
- Designed a risk/reward survival loop with a decaying health pool that forces players to balance speed and health
- Built a rooftop-based environment using verticality and obstacle placement to test each traversal mechanic

SKILLS

- Languages: Python, Java, JavaScript, C, C++, C#, Rust, HTML, CSS
- Frameworks & Libraries: React, Svelte, PyTorch, TensorFlow, Pandas, NumPy, LangChain
- Tools & Platforms: Unity, Unreal Engine, Git/GitHub, Firebase, Supabase, ChromaDB, Xcode, Terminal

AWARDS

- LMU Arrupe 4-Year Scholarship
- Dean's List (2023, 2024, 2025, 2026)